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#include <iostream> #include <cmath> using namespace std;
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int recurrenceT(int n) {  
    if (n == 1) {  
        return 1;  
    }  
    return 2 * recurrenceT(n / 2) + n; }  

```

```
int main() {  
    cout << "Substitution Method Example" << endl;  
    cout << "Recurrence:  $T(n) = 2T(n/2) + n$ " << endl;  
    cout << "Solution:  $T(n) = O(n \log n)$ " << endl;  
    cout << endl;  
  
    cout << "Testing values of  $T(n)$ :" << endl;  
    for (int n = 1; n <= 16; n *= 2) {  
        cout << "T(" << n << ") = " << recurrenceT(n) << endl;  
    }  
  
    cout << endl;  
    cout << "Verification with  $n \cdot \log(n)$ :" << endl;  
    for (int n = 1; n <= 16; n *= 2) {  
        double expected = n * log2(n);  
        cout << "n = " << n << ",  $n \cdot \log(n)$  = " << expected << ",  $T(n)$  = " << recurrenceT(n) << endl;  
    }  
  
    return 0; }  

```